CS341 · COMPUTER GRAPHICS

Midterm 451 Practice

30 marks · 1 hour

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01 Questions

Q1 SHORT ANSWER · 5 PTS

1. Define computer graphics?

2. What is the difference between emissive displays and nonemissive displays?

Q2 MCQ · 5 PTS

1. In raster scan systems, the picture information is stored in _____

a) Display list b) Refresh buffer c) Vector file d) None of all

2. Raster images are more commonly called:

a) Pix map b) Bitmap c) Both 1 & 2 d) None of these

3. The quality of the image depends on:

a) Number of pixels used by the image b) Number of lines used by the image c) Number of photos used by the image d) All the above

4. The frequency at which a picture is redrawn on the screen is referred to as the:

a) Beam-penetration b) Shadow-mask c) Resolution d) Refresh rate

5. One of the applications of computer graphics:

a) Education & Training b) Computer Art c) Paint programs d) All of above

Q2 MCQ · CONT.

6. The graphics can be:

a) Drawing b) Photograph, movies c) Simulation d) All of these

7. Higher the number of pixels, _____ the image quality.

a) Bad b) Better c) Smaller d) None of above

8. Each screen point is referred to as _____

a) Resolution b) Pixel c) Persistence d) Dot pitch

9. Rays emitted by cathode ray tube are:

a) Light b) Radiations c) Signals d) Electrons

10. Active-matrix display included:

a) Two plates of glass b) Each pixel on the screen is controlled by its own transistor c) A transistor controls a whole row or column of pixels d) The screen needs less power

Q3 LINEAR COMBINATIONS · 2.5 PTS

Complete the following:

a) Point – Point =

b) Point + Vector =

c) Vector + Vector =

d) Scalar * Vector =

e) Scalar * Vector + Scalar * Vector =

Q4 SHORT ANSWER · 2 PTS

1. Write four of the hard-copy devices.

2. Write four of the video display devices.

Q5 BRESENHAM · 13 PTS

1. Based on the Bresenham algorithm, calculate the points between the starting point (1, 1) and the ending point (5, 9). Then, draw the line using these points.

2. Given a circle radius r = 7, demonstrate the Bresenham circle drawing algorithm by determining positions along the circle octant in the first quadrant from x = 0 to x = y.